

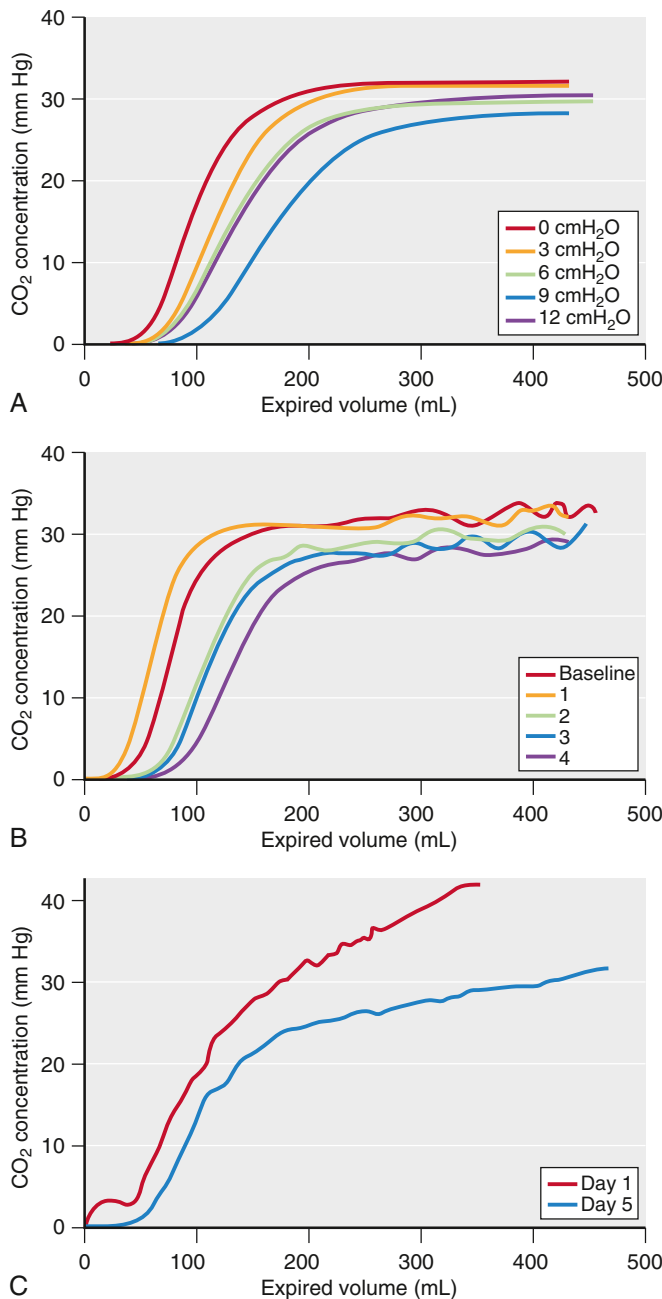


Fig. 37.14 Representative single-breath volume capnograms, illustrating various abnormalities in ventilation or perfusion. (A) Alterations in phases II and III with corresponding changes in positive end-expiratory pressure (0, 3, 6, 9, and 12 cmH₂O) during positive pressure ventilation; (B) alterations in phases II and III with corresponding changes in pulmonary perfusion (increasing numbers correspond to decreasing pulmonary blood flow); (C) pronounced positive slope of phase III during acute bronchospasm (Day 1). Following resolution (Day 5), there is a noticeable reduction in the slope phase III. CO₂, carbon dioxide. (Modified from Thompson JE, Jaffe MB. Capnographic waveforms in the mechanically ventilated patient. *Respir Care*. 2005;50:100–108.)

and compensate for the low $\dot{V}/\dot{Q}$ regions (Fig. 37.15). Consequently, $\dot{V}/\dot{Q}$ mismatch results in hypoxemia.

Right-to-left shunt is the amount of blood that flows from the pulmonary artery to the systemic arterial circulation without undergoing pulmonary gas exchange. It represents an extreme case of $\dot{V}/\dot{Q}$ mismatch in which the ratio equals zero and the end-capillary gas partial pressures are equal to

the values found in mixed venous blood. In healthy, awake, spontaneously breathing subjects, intrapulmonary shunt is negligible,¹⁹⁵ and a small (<1% of cardiac output) extrapulmonary shunt results from drainage of the bronchial and Thebesian veins into the arterial side of the circulation.¹⁹⁶ During general anesthesia, a right-to-left shunt can develop as a result of atelectasis.^{197,198} Right-to-left shunt can also be seen in pathologic conditions such as pneumonia and acute lung injury. The effect of the shunt on P_aO_2 is a function of the magnitude of the shunt, F_iO_2 , and the cardiac output (Fig. 37.16). Importantly, increases in F_iO_2 have a small effect on P_aO_2 in the presence of large true right-to-left shunt (see Fig. 37.16).

The traditional method to estimate flow through shunting regions ($\dot{Q}_s$) as a fraction of the total cardiac output ($\dot{Q}_T$) is based on the modeling of the lung as a three-compartment system (Fig. 37.17).¹⁹⁹ The three compartments represent (1) lung regions receiving both ventilation and perfusion, (2) a shunt compartment ($\dot{V}/\dot{Q} = 0$), and (3) a dead space compartment with ventilation but no perfusion ($\dot{V}/\dot{Q} = \text{infinity}$). The application of the concept of mass balance to this model yields the shunt fraction ($\dot{Q}_s/\dot{Q}_T$) expression

$$\dot{Q}_s / \dot{Q}_T = \frac{(Cc'O_2 - CaO_2)}{(Cc'O_2 - CvO_2)} \quad (37.16)$$

where $\dot{Q}_s$ is the shunted blood flow, and $Cc'O_2$ is the end-capillary O₂ content. $Cc'O_2$ is calculated using Eq. 37.1, assuming equilibration between the end-capillary blood and alveolar gas. The P_aO_2 is calculated using the ideal alveolar gas equation:

$$P_aO_2 = F_iO_2 \cdot (P_{\text{atm}} - P_{H_2O}) - P_aCO_2 / R \quad (37.17)$$

where F_iO_2 is the inspired O₂ fraction, P_{atm} is barometric pressure (typically assumed to be the pressure of one standard atmosphere at sea level, 760 mm Hg), and P_{H_2O} is partial pressure of water vapor (47 mm Hg at 37°C). R , the respiratory quotient, is the ratio of pulmonary CO₂ elimination to O₂ uptake ($R = \dot{V}CO_2/\dot{V}O_2$), which equals the value of 0.8 under normal diet and metabolic conditions.

The three-compartment model is a simplification of the real lung. As a result, under conditions of F_iO_2 less than 100%, $\dot{Q}_s/\dot{Q}_T$ computed with Eq. 37.16 represents a combination of all factors producing hypoxemia, predominantly true right-to-left shunt and $\dot{V}/\dot{Q}$ mismatch. Under such conditions, $\dot{Q}_s/\dot{Q}_T$ is referred to as venous admixture. When F_iO_2 is 100%, the effects of $\dot{V}/\dot{Q}$ heterogeneity on O₂ exchange are eliminated and the equation exclusively yields the right-to-left shunt fraction.²⁰⁰ The administration of 100% O₂ (i.e., F_iO_2 of 1.0) can lead to the development of absorption atelectasis in compartments with very low $\dot{V}/\dot{Q}$, thus leading to an increase in true right-to-left shunt.²⁰⁰

If the assumptions are made that end-capillary blood has an O₂ saturation of 100% and the O₂ content of blood is predominantly determined by Hb saturation, the shunt fraction equation can be simplified to

$$\dot{Q}_s / \dot{Q}_T = \frac{(1 - S_aO_2)}{(1 - S\bar{v}O_2)} \quad (37.18)$$

where S_aO_2 is arterial saturation, and $S\bar{v}O_2$ is mixed venous saturation.

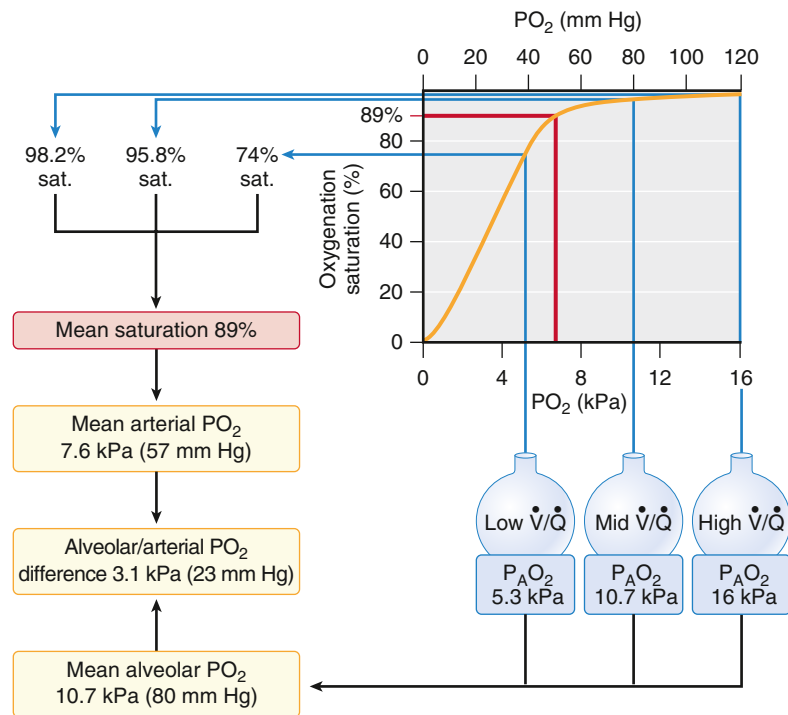


Fig. 37.15 Effect of different ventilation-perfusion ($\dot{V}/\dot{Q}$) regions on partial pressure of arterial oxygen (P_aO_2). Because of the shape of the oxyhemoglobin dissociation curve, regions of lung with low $\dot{V}/\dot{Q}$ have a disproportionately greater effect on lowering the arterial saturation than do regions of lung with normal or higher $\dot{V}/\dot{Q}$ to increase P_aO_2 . Assuming equal blood flow, the average partial pressure of alveolar oxygen (P_AO_2) of the $\dot{V}/\dot{Q}$ regions represented in the illustration is 10.7 kPa (80 mm Hg). Based on the oxyhemoglobin dissociation curve, however, the average saturation of the arterial blood is 89%, producing an average P_aO_2 equal to 7.6 kPa (57 mm Hg). PO_2 , partial pressure of oxygen. (Modified from Lumb AB. *Nunn's Applied Respiratory Physiology*. 6th ed. Philadelphia: Elsevier/Butterworth Heinemann; 2005.)

Other indices of oxygenation

The calculation of the shunt fraction is a fundamental measure of O₂ exchange impairment. However, a pulmonary artery catheter is required for the measurement of $\bar{S}\bar{V}O_2$. For this reason, indices of oxygenation that rely on less invasive measures have been developed. An ideal index should reflect the efficiency of oxygenation, change with changes in lung function, remain constant with changes in extrapulmonary conditions (such as F_{IO_2}), and provide clinically useful diagnostic and prognostic information.²⁰¹ Although P_aO_2 certainly reflects arterial blood oxygenation, it is limited in providing a measure of the magnitude of the O₂ exchange deficiency because of its dependence on the F_{IO_2} and to the nonlinear relationship between P_aO_2 and blood O₂ content.²⁰¹ As a consequence, several indices based on the P_aO_2 that account for the F_{IO_2} or the P_AO_2 have been developed: the alveolar–arterial partial pressure gradient of O₂ ($[A-a]PO_2$), the respiratory index ($[A-a]PO_2/P_aO_2$), the arterial–alveolar ratio of O₂ partial pressures (P_aO_2/P_AO_2), and the ratio between P_aO_2 and F_{IO_2} (P_aO_2/F_{IO_2}).²⁰¹

The advantages of indices based on P_aO_2 are that their calculations are fairly simple and require only arterial blood sampling. However, an important limitation is that these indices vary with the F_{IO_2} , P_aCO_2 , Hb, and O₂ consumption ($\dot{V}O_2$).^{202,203} Consequently, changes in those variables can lead to changes in the values of the indices in the absence of any change in lung gas exchange function. Another limitation is the failure of these indices to account for changes in $\dot{V}/\dot{Q}$ matching resulting from changes in the F_{IO_2} . In addition, indices that use P_AO_2 rely on the assumptions of

the alveolar gas equation, including $P_ACO_2 = P_aCO_2$, which may not hold true in pathologic conditions.

An early index developed to assess oxygenation that used only peripheral blood sampling was the (A-a)PO₂.²⁰⁴ The (A-a)PO₂ can help to differentiate $\dot{V}/\dot{Q}$ mismatch, shunt, and diffusion limitation from hypoventilation and low F_{IO_2} as causes of hypoxemia. This is because the (A-a)PO₂ is unchanged under conditions of decreased F_{IO_2} and hypoventilation but increased with $\dot{V}/\dot{Q}$ mismatch, shunt, and diffusion limitation.

The A-a gradient of O₂ is calculated as

$$(A-a)PO_2 = P_AO_2 - P_aO_2 \quad (37.19)$$

The normal value of the (A-a)PO₂ is less than 10 mm Hg in young adults breathing room air, and it increases with age and the administration of supplemental O₂. The change in (A-a)PO₂ with age can be estimated by²⁰⁵

$$(A-a)PO_2 = 0.21 \cdot (\text{age} + 2.5) \quad (37.20)$$

The (A-a)PO₂ varies significantly with the F_{IO_2} such that the application of supplemental O₂ can lead to an increase in the (A-a)PO₂ independent of pulmonary disease.²⁰² Like the (A-a)PO₂, the respiratory index $(A-a)PO_2/P_aO_2$ and the P_aO_2/P_AO_2 are sensitive to F_{IO_2} , especially in the presence of a high degree of $\dot{V}/\dot{Q}$ mismatch, such as during acute respiratory distress syndrome (ARDS), and less so in the absence of true shunt or when low $\dot{V}/\dot{Q}$ regions are minimal (e.g., in healthy lung or pulmonary embolus).²⁰⁶

Unlike the other indices based on partial pressures, the P_aO_2/F_{IO_2} ratio does not use the P_AO_2 and its associated assumptions. This index is more stable, particularly under

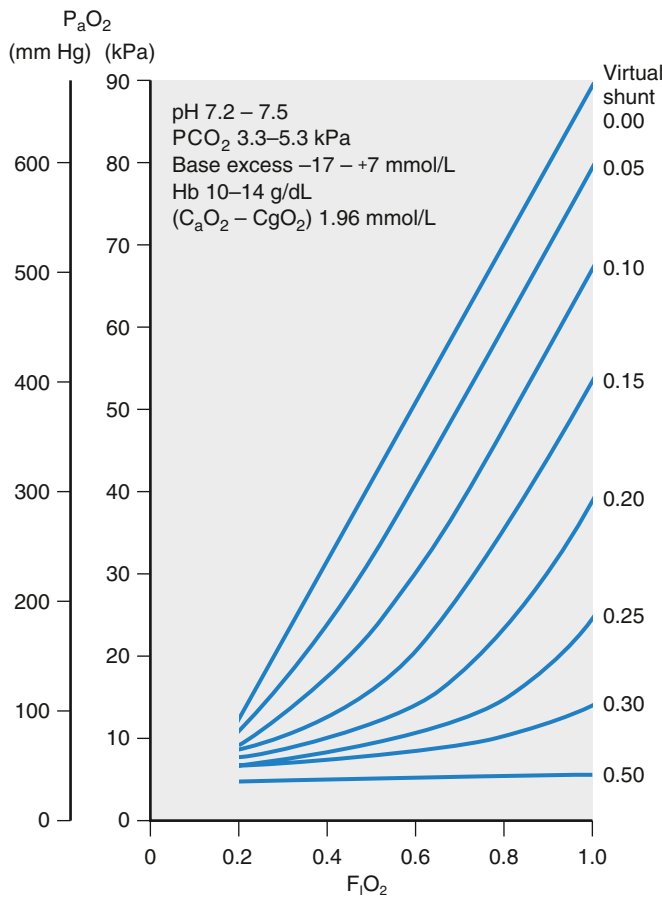


Fig. 37.16 Relationship between partial pressure of arterial oxygen (P_{aO_2}) and fraction of inspired oxygen (F_{iO_2}) for different values of shunt fraction computed for normal values of barometric pressure, pH, partial pressure of carbon dioxide (PCO_2), base excess, hemoglobin (Hb) concentration, and arterial-mixed venous oxygen content difference ($C_{aO_2} - C_{vO_2}$). Note the small increase in P_{aO_2} with F_{iO_2} in the presence of large shunt. This graph corresponds to the so-called isoshunt diagram presented by Lawler and Nunn. (From Welsby PD, Earis JE. Some high pitched thoughts on chest examination. *Postgrad Med J*. 2001;77:617–620.)

conditions relevant to ARDS such as P_{aO_2} lower than 100 mm Hg in the setting of an F_{iO_2} greater than 0.5.^{202,203} P_{aO_2}/F_{iO_2} values are part of the diagnostic criteria for ARDS and have been correlated with outcomes in these patients.²⁰⁷ When repeated arterial blood sampling is not possible, the S_{pO_2}/F_{iO_2} (also referred to as SF) ratio may be useful. SF ratio values have shown good correlation with P_{aO_2}/F_{iO_2} values in both adults and children with respiratory failure who have S_{pO_2} values in the range of 80% to 97%.^{208,209}

The P_{aO_2}/F_{iO_2} ratio does not characterize factors associated with the severity of respiratory failure such as ventilator settings or respiratory system mechanics.^{210–212} Another drawback of the P_{aO_2}/F_{iO_2} ratio is its dependence on PEEP or mean airway pressure, because increases in these variables result in lung recruitment and improved oxygenation. As a consequence, the oxygenation index (OI) was proposed as a more robust alternative to the P_{aO_2}/F_{iO_2} ratio²¹³:

$$OI = 100 \times \frac{F_{iO_2} \bar{P}_{ao}}{P_{aO_2}} \quad (37.21)$$

where $\bar{P}_{ao}$ denotes the mean airway pressure. The OI has been used as a prognostic indicator in pediatric patients with acute hypoxemic respiratory failure,²¹⁴ as a predictor of mortality in adult ARDS,^{212,215} and as an index of enhanced lung recruitment during oscillatory ventilation.²¹⁶ An alternative “oxygenation saturation index” (OSI) has also been recently proposed:

$$OSI = 100 \times \frac{F_{iO_2} \bar{P}_{ao}}{S_{pO_2}} \quad (37.22)$$

Similar to the pulse oximetric saturation S_{pO_2}/F_{iO_2} (SF) ratio, the OSI eliminates the need for invasive arterial blood gas sampling by using pulse oximetry. When computed on the day of ARDS diagnosis, it correlates well with the OI and is associated with increased mortality and fewer ventilator-free days.²¹⁰

Unlike P_{aO_2} , P_{aCO_2} can be maintained in the normal range in the presence of significant $\dot{V}/\dot{Q}$ mismatching. The reason is that the CO_2 content curve has no plateau. In this way, compensatory hyperventilation can be used to reduce elevations in P_{aCO_2} .²¹⁷ The dependence of P_{aCO_2} on CO_2 elimination ($\dot{V}CO_2$) and alveolar ventilation ($\dot{V}_A$), with k in the equation as a conversion factor equal to 0.863, is expressed by

$$P_{aCO_2} = k \times (\dot{V}CO_2 / \dot{V}_A) \quad (37.23)$$

In steady-state conditions (constant $\dot{V}CO_2$), P_{aCO_2} changes inversely with the alveolar ventilation. The traditional measure of ventilation inefficiency has been the fraction of dead space to tidal volume (V_D/V_T) as calculated using the Bohr equation (Eq. 37.15) and the assumption that the arterial and alveolar partial pressures of CO_2 are equivalent (i.e., $P_ACO_2 = P_{aCO_2}$). The dead space fraction determined in this manner includes alveolar (i.e., nonperfused alveoli) and anatomic dead space, as well as regions of high $\dot{V}/\dot{Q}$ ratio. As with the discussion on shunt, the dead space fraction characterizes not only real anatomic dead space, but also areas of inefficient ventilation (see Fig. 37.17).

MEASUREMENT PRINCIPLES

P_{aO_2} is measured using a Clark electrode. The electrode consists of a cathode (either platinum or gold) and an anode in an electrolytic bath, surrounded by a thin O_2 permeable membrane. The electrode is inserted into the sample, and O_2 diffuses through the membrane and is reduced by the cathode, generating a current. The current is proportional to the PO_2 in the sample.²¹⁸ The PCO_2 electrode measures the change in pH brought about by the equilibration of blood with a bicarbonate solution.²¹⁹

Effect of Temperature

The solubility of CO_2 and O_2 in blood is affected by temperature. At lower temperatures, solubility is increased, leading to a reduction in partial pressure. Blood gas analyzers measure gas partial pressures at 37°C. As blood from a hypothermic patient is warmed to 37°C by the analyzer, CO_2 and O_2 will come out of solution, leading to P_{aCO_2} and P_{aO_2} higher than those present in the patient. Blood gas analyzers make use of algorithms to correct the reported values to the patient's actual temperature.²²⁰ The temperature effects on blood gas analysis are most relevant during management of

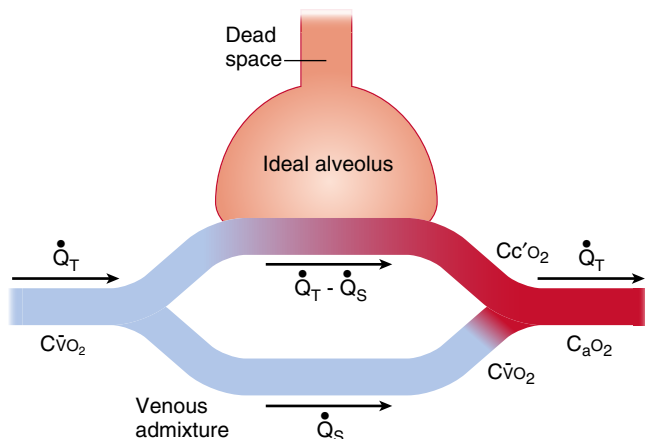


Fig. 37.17 Three-compartment model of gas exchange. The lung is represented as three functional units: alveolar dead space, “ideal” alveoli, and venous admixture (shunt). Note that the model doesn’t differentiate between true shunt or true dead space and components of shunt and dead space that are caused by ventilation–perfusion ($\dot{V}/\dot{Q}$) mismatch. C_aO_2 , arterial oxygen content; $C_c'O_2$, oxygen content of end pulmonary capillary blood; C_VO_2 , mixed venous oxygen content; $\dot{Q}_S$, shunted blood flow; $\dot{Q}_T$, total blood flow. (From Lumb AB. *Nunn’s Applied Respiratory Physiology*, 6th ed. Philadelphia: Elsevier/Butterworth Heinemann; 2005.)

hypothermic patients, such as during hypothermic cardiopulmonary bypass (HCPB) or deep hypothermic circulatory arrest (DHCA).

Two strategies have been developed for the management of arterial gas partial pressures under these conditions: alpha-stat and pH-stat. During alpha-stat management, blood gas measurements obtained after the analyzer warms the sample to 37°C are used for acid–base and gas exchange management. Potential benefits of alpha-stat management include the preservation of cerebral autoregulation and the maintenance of protein function.²²¹ During pH-stat management, measurements are corrected to the patient’s temperature before they are used for acid–base and gas exchange management. Because patients are hypothermic, the P_aO_2 and P_aCO_2 at the patient’s temperature are lower and the pH higher than those measured in the analyzer at 37°C. CO_2 is usually added to the oxygenator to maintain the temperature-corrected P_aCO_2 and pH at normothermic values. The theoretical benefits of pH-stat management are increased cerebral vasodilation leading to more homogeneous cerebral cooling.²²²

Studies have compared the two strategies with respect to outcomes with varying results.^{223–230} In general, clinical studies support the use of pH-stat management in pediatric cardiac surgical procedures during HCPB or DHCA.^{225,229,231} In adult patients, the use of alpha-stat management is supported for HCPB.²³² For procedures involving DHCA in adults, the use of pH-stat management has been suggested to increase the rate and uniformity of cooling and alpha-stat management during rewarming.²³³

LIMITATIONS AND SOURCES OF ERRORS

Proper handling of arterial blood gas samples is important to prevent errors. Two common sources of error are delays in sample analysis and the presence of air in the sampling syringe.²³⁴ A delay in sample analysis by 20 minutes at

room temperature or at 4°C can lead to a decline in P_aO_2 .²³⁴ The decline is attributed to the metabolic activity of leukocytes and is not observed in samples placed on ice. The presence of an air bubble in the syringe can lead to a change in the measured P_aO_2 toward the PO_2 of the bubble and decline in P_aCO_2 .²³⁴

Monitoring Lung Flows, Volumes, and Pressures

As an organ of gas exchange, the human respiratory system relies on convective and diffusive gas transport processes for VO_2 and elimination of CO_2 . The transport of air to the alveoli and alveolar gas to the environment requires the creation of a pressure gradient, which results in volume changes in the elastic components of the respiratory system, flow in the airways, velocity of moving tissues, and acceleration of air and tissues. The lung consists of a complex branching network of airway segments and viscoelastic tissues that, during the processes of spontaneous breathing or mechanical ventilation, gives rise to tremendous variations in gas velocities and flow regimens. The movement of gas flow ($\dot{V}$) into and out of the lungs requires pressure (P) to overcome the resistive (P_R), elastic (P_E), and at times inertial (P_I) forces offered by the airway tree, parenchymal tissues, and chest wall:

$$P = P_R + P_E + P_I \quad (37.24)$$

Eq. 37.24 can be used to describe the mechanical behavior of the total respiratory system, the lungs alone, or the chest wall, depending on whether P refers to airway pressure relative to atmosphere (transrespiratory pressure), airway pressure relative to pleural pressure (transpulmonary pressure), or pleural pressure alone.^{235–237}

DYNAMIC RESPIRATORY MECHANICS

As a first approximation to Eq. 37.24, the mechanical behavior of the respiratory system during breathing or ventilation can be described according to the simple equation of motion that encompasses its resistive (R), elastic (E), and inertial (I) properties^{235,238}:

$$P = R\dot{V} + E\ddot{V} + I\ddot{V} + P_0 \quad (37.25)$$

where V is volume, $\ddot{V}$ denotes volume acceleration (i.e., the first time-derivative of flow or the second time-derivative of volume), and P_0 is the distending pressure at end expiration. The coefficients R , E , and I in Eq. 37.25 may refer to the mechanical properties of the total respiratory system (rs), the lungs alone (L), or the chest wall (cw), depending on whether P refers to transrespiratory pressure, transpulmonary pressure, or intrapleural pressure, respectively.²³⁷ The resistive properties of the respiratory system (R_{rs}) are generally assumed to arise from viscous and turbulent losses associated with gas flowing through the airway tree and the deformation of parenchymal and chest wall tissues. Hence, airway resistance may reflect airway caliber.²³⁹ Provided that variations in flow are small, resistive pressure losses are assumed to be linearly related to flow according to the first term of Eq. 37.25. With more rapid flow rates, as may be seen during exercise or forced vital capacity maneuvers,

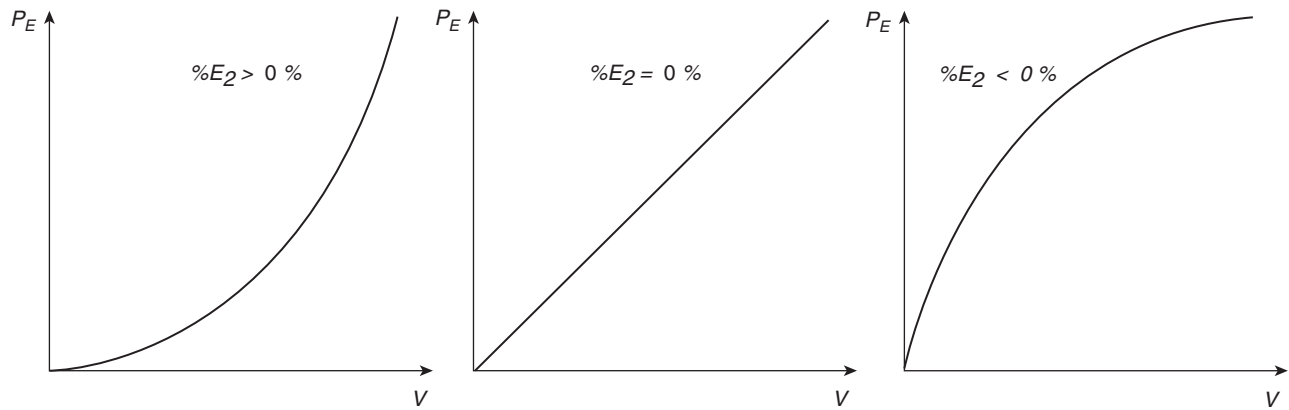


Fig. 37.18 Example elastic pressure (P_E) versus volume (V) curves and corresponding percentage of volume-dependent respiratory system elastance ($\%E_2$) values. (Modified from D'Antini D, Huhle R, Herrmann J, et al. Respiratory system mechanics during low versus high positive end-expiratory pressure in open abdominal surgery: a substudy of PROVHILO randomized controlled trial. *Anesth Analg*. 2018;126:143–149.)

resistive losses vary nonlinearly with flow, which may more accurately be described according to²⁴⁰

$$P_R = K_1 \dot{V} + K_2 \dot{V}^2 \quad (37.26)$$

where K_1 and K_2 are empirically determined constants. Additional energy losses arise from the tension within the alveolar surface film,²⁴¹ friction within the pleural space, and various tissue components of the parenchyma and chest wall,²³⁶ as well as cross-bridge cycling of contractile elements within the airway walls and lung tissues.^{242,243} Such losses are collectively termed “tissue resistance.”^{244,245} If these tissue resistive losses are assumed to be proportional to flow according to Eq. 37.25, one will observe that they vary inversely with breathing frequency,^{242,246} a phenomenon commonly associated with viscoelastic materials.²³⁶ In adult patients, lung tissue resistance comprises about 60% of subglottal total lung resistance at typical breathing rates.²⁴⁷ During volume-cycled ventilation with an end-inspiratory pause, resistance can quickly be estimated from the difference between the peak (P_{peak}) and plateau (P_{plat}) airway pressures (i.e., the resistive pressure loss P_R) divided by the inspiratory flow ($\dot{V}_I$) immediately before the end-inspiratory pause:

$$R = \frac{P_{\text{peak}} - P_{\text{plat}}}{\dot{V}_I} = \frac{P_R}{\dot{V}_I} \quad (37.27)$$

Thus, for a specified flow rate, alterations in P_R may reflect changes in airway caliber, as may occur during an intraoperative asthma exacerbation (Fig. 37.18, center panel), or obstruction within the ETT or breathing circuit.

Elastic pressures arise from the recoil of the lungs and chest wall when their normal anatomic configurations are deformed, either by contraction of the diaphragm and intercostal muscles, or by some external force such as a ventilator.²⁴⁸ Elastance is defined as the change in distending pressure (transrespiratory, transpulmonary, or intrapleural) for a given change in volume. *Dynamic elastance* refers to the change in elastic (i.e., nonresistive) distending pressure per unit volume during the process of breathing or ventilation[†]. Elastance of the total respiratory

system (E_{rs}) comprises both lung (E_L) and chest wall (E_{cw}) elastances:

$$E_{rs} = E_L + E_{cw} \quad (37.28)$$

In clinical settings, elastance is usually expressed as its reciprocal, compliance (i.e., change in volume per unit change in pressure). This leads to an alternate form of Eq. 37.28, expressing the relationships of the compliant components of the total respiratory system (C_{rs}), lungs (C_L), and chest wall (C_{cw}):

$$\frac{1}{C_{rs}} = \frac{1}{C_L} + \frac{1}{C_{cw}} \quad (37.29)$$

Factors that increase total respiratory or lung elastance (or likewise reduce the corresponding total respiratory or lung compliance) include consolidation, pulmonary edema, pneumothorax, atelectasis, interstitial disease, pneumonectomy or surgical resection, lung overdistention, and mainstem intubation. Lung compliance generally increases in emphysema. Factors that reduce chest wall compliance are abdominal distention, abdominal compartment syndrome, chest wall edema, thoracic deformity, muscle tone, and extensive thoracic or abdominal scar (e.g., from burns); C_{cw} is reduced with muscle relaxation and flail chest.

In some situations, the magnitude of V_T , relative to the functional lung size, may be such that the linear dependence of P_E on V (Eqs. 37.24 and 37.25) is no longer an accurate description of the pressure–volume (PV) relationship of the lungs and/or chest wall. In these cases, elastic pressure may be more accurately described by a quadratic relationship with volume^{249,250}:

$$P_E = E_1 V + E_2 V^2 \quad (37.30)$$

in which E_1 and E_2 denote the so-called volume-independent and volume-dependent components of elastance, respectively, since P_E will be reduced to $E_1 V$ when E_2 is negligible (Eq. 37.23). An advantage of using Eq. 37.30 is that it readily allows for the determination of the percentage of volume-dependent elastance contributing to P_E ²⁵¹:

$$E_2 = \frac{E_2 V_T}{E_1 + E_2 V_T} \times 100 \quad (37.31)$$

[†]Certain conventions of nomenclature incorrectly define dynamic elastance as the peak airway pressure divided by tidal volume. However, such a definition of dynamic elastance includes both resistive and elastic pressures, and thus is not an index of purely elastic processes.

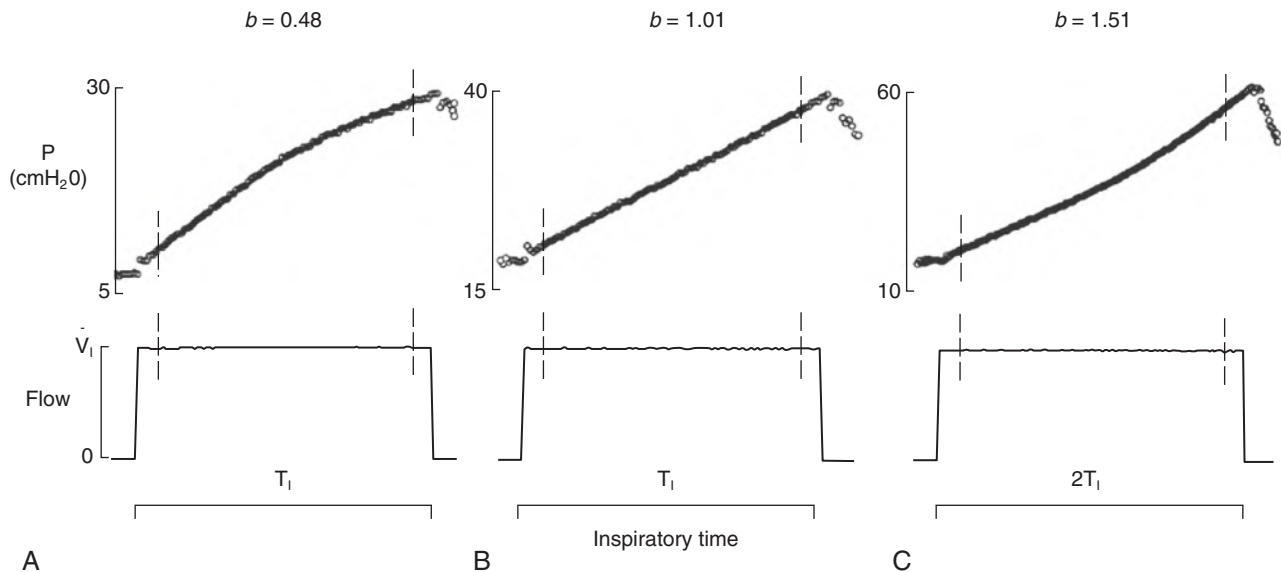


Fig. 37.19 Conceptual example of the stress index during constant inspiratory flow levels of $\dot{V}_i$ over durations of T_i and $2T_i$, assuming that the power law ($P = at^b + c$) can describe the pressure–time (P–t) curve. (A) $b < 1.0$, yielding a convex P–t curve, suggesting the dominance of intratidal recruitment; (B) $b \sim 1.0$, yielding a straight P–t line, suggesting minimal recruitment or overdistention; and (C) $b > 1.0$, yielding a concave P–t curve, suggesting the dominance of parenchymal overdistention. (Modified from Ranieri VM, Zhang H, Mascia L, et al. Pressure–time curve predicts minimally injurious ventilatory strategy in an isolated rat lung model. *Anesthesiology*. 2000;93:1320–1328.)

Since the $\%E_2$ index quantifies the deviation of linearity between P_E and V , it may provide quantitative information on the degree of intratidal recruitment or overdistention occurring during a mechanical breath.²⁵² For example, positive values of $\%E_2$ (see Fig. 37.18A) indicate a convex P_E – V relationship, with probable overdistention occurring for $\%E_2$ greater than 30%.²⁵¹ By contrast, negative values of $\%E_2$ (see Fig. 37.18C) indicate concave P_E – V relationship, with likely intratidal recruitment occurring during the breath.²⁵³

For the special case of constant inspiratory flow during volume-cycled ventilation, the curvature of the inspiratory pressure–time profile can also be used to infer the processes of intratidal recruitment and overdistention. Raneiri and associates²⁵⁴ described this relationship using a simple power law expression:

$$P = at^b + c \quad (37.32)$$

where, similar to Eq. 37.24, P may refer to either transrespiratory or transpulmonary pressure, and t is time. The constants a , b , and c can be estimated from the inspiratory pressure profile using various nonlinear regression techniques.²⁵⁵ The exponent b is referred to as the “stress index,” and its estimated value characterizes the convexity ($b < 1.0$), concavity ($b > 1.0$), or linearity ($b = 1.0$) of the inspiratory pressure with time (Fig. 37.19). Similar to $\%E_2$, b may also provide some insight into the relative contributions of intratidal recruitment ($b < 1.0$) and parenchymal overdistention ($b > 1.0$) to inspiratory pressure, or when these processes may be appropriately balanced ($b \sim 1$). However, as noted, the stress index is only interpretable during volume-cycled ventilation with constant inspiratory flows, in contrast to the $\%E_2$ index, which can be used during other ventilatory modalities.²⁵⁶ Moreover, the interpretation of the stress index may be obscured during certain clinical situations, such as abdominal insufflation, abdominal compartment syndrome, or pleural effusion.^{257–259}

Similar to Eq. 37.27 to estimate R_{rs} , linear approximations to respiratory system elastance (E_{rs}) can be easily determined during volume-cycled ventilation for periods of zero flow (i.e., during an end-inspiratory pause, when resistive pressures are zero) as the difference between P_{plat} and PEEP (i.e., the elastic pressure P_E) divided by the V_T :

$$E_{rs} = \frac{P_{plat} - PEEP}{V_T} = \frac{P_E}{V_T} \quad (37.33)$$

Since the value of E_{rs} in Eq. 37.33 is obtained during zero flow, it is often referred to as “static elastance.” The corresponding static C_{rs} is usually between 50 and 100 mL/cmH₂O in normal mechanically ventilated lungs. Alterations in P_E for a specified V_T may be observed during processes that alter lung or chest wall elastance, such as a pneumothorax during thoracotomy or abdominal insufflation during laparoscopic surgical procedures (Fig. 37.20, right panel). In contrast to static E_{rs} , dynamic E_{rs} is estimated during periods of nonzero flow by using multiple linear regression techniques (see later), and is higher than static elastance because of viscoelasticity²³⁸ and gas redistribution.²⁶⁰ Both the apparent static or dynamic E_{rs} will decrease following a recruitment maneuver because of an increase in aerated or functional lung volume. However, E_{rs} may also be increased due to *strain stiffening* of the lung tissues,²⁶¹ in which collagen fibers within the extracellular matrix of the parenchyma become progressively recruited with increasing strain. Recruitment of stiff collagen fibers results in their dominance of load-bearing within the lung, in contrast to the more stretchable elastin fibers that are the main load-bearing elements during low levels of tissue strain.²⁶² Such behavior may allow for titration of PEEP to achieve an optimal balance between intratidal recruitment and parenchymal overdistention – that is, the PEEP for which E_{rs} is minimized or, equivalently, C_{rs} is maximized.^{249,250,263–266}

The clinical utility of such PEEP titration in patients with ARDS is still not entirely established.^{265,267} Finally, inertial

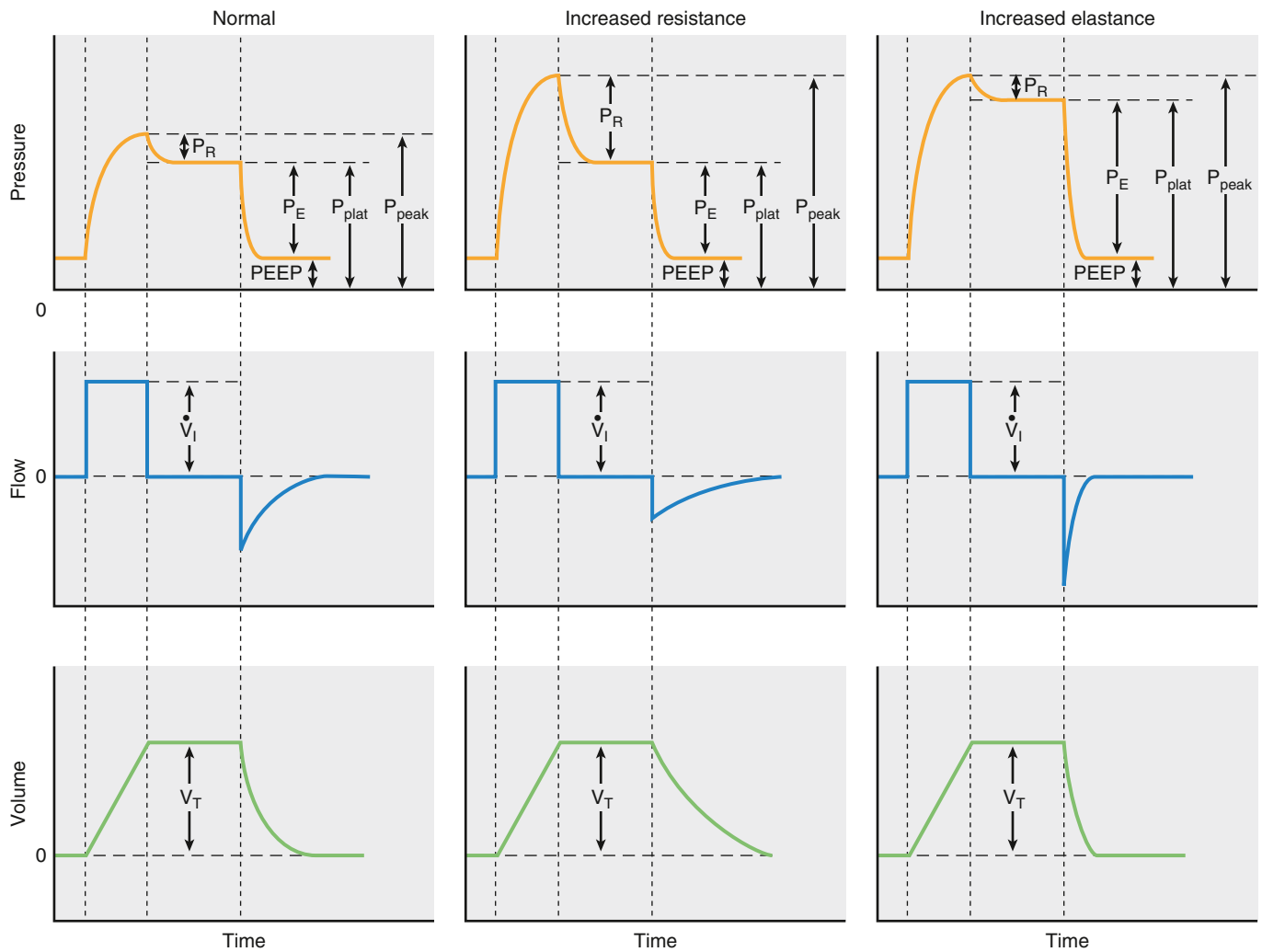


Fig. 37.20 Airway pressure, flow, and volume waveforms during constant-flow volume-cycled ventilation with an end-inspiratory pause. The components of the peak inspiratory pressure (P_{peak}) and the plateau pressure (P_{plat}) are presented: P_R , resistive pressure; P_E , elastic pressure; and $PEEP$, positive end-expiratory pressure. The same peak flow ($\dot{V}_I$) and tidal volume (V_T) are delivered by the mechanical ventilator in each case. Leftmost panels represent waveforms in a healthy patient. There is a small resistive pressure component, with most of the peak pressure composed of elastic pressure. Expiratory flow decreases to zero before the start of the next inspiration, indicating appropriate I:E ratio and the absence of slowly emptying lung regions. Volume, as the integral of flow, also returns to zero before the initiation of the next breath. Middle panels represent waveforms in a patient with an increase in airways resistance. There is a significant increase in the resistive component P_R , and consequently P_{peak} . The elastic component is unchanged, as reflected in identical P_{plat} as compared with the healthy condition. Because increased resistance slows the process of lung emptying, the expiratory phase of the flow and volume curves now take longer to reach the zero. In several cases, the zero will not be reached, and a change in the I:E ratio will be necessary to avoid volume trapping and auto-PEEP. Rightmost panels represent a subject with increased P_E , as may occur during laparoscopic insufflation or pneumothorax. There is an increase in P_{plat} consistent with the increased elastance of the respiratory system, and a fast emptying of the lungs during exhalation.

pressures are associated with the kinetic energy of accelerating the gas column in the central airways, as well as the motion of the respiratory tissues.²⁶⁸ Inertial pressures are typically expressed as the product of a lumped “inertia” parameter and volume acceleration (Eq. 37.25). Inertia is usually not a significant contributor to the apparent airway pressure or the work of breathing, except during sudden changes in air flow, as may occur during inspiratory step profiles or various modalities of high-frequency ventilation (HFV).^{269–271}

Eqs. 37.27 and 37.33 cannot be used to estimate R_{rs} and E_{rs} if an end-inspiratory pause is not present in the ventilator waveform. This is because one cannot easily separate the resistive and elastic pressures by visual inspection alone. For example, during pressure-controlled

ventilation (PCV), the airway opening is exposed to a constant inflation pressure during inspiration (Fig. 37.21). Airway flow and V_T during PCV are determined not by the ventilator, but rather by the gradient between airway and alveolar pressures. Thus, a more robust method of estimating respiratory R_{rs} and E_{rs} involves multiple linear regression of the coefficients in Eq. 37.25 on sampled flow and pressure data.^{272–274} Such numerical approaches are present in commercially available respiratory mechanics monitors, and they can be used on almost all ventilator waveforms. Moreover, estimates of mechanical properties based on linear regression are not restricted to volume- or time-cycled waveforms with an end-inspiratory pause. One can separately estimate R_{rs} and E_{rs} during the inspiratory and expiratory phases, which may be different in

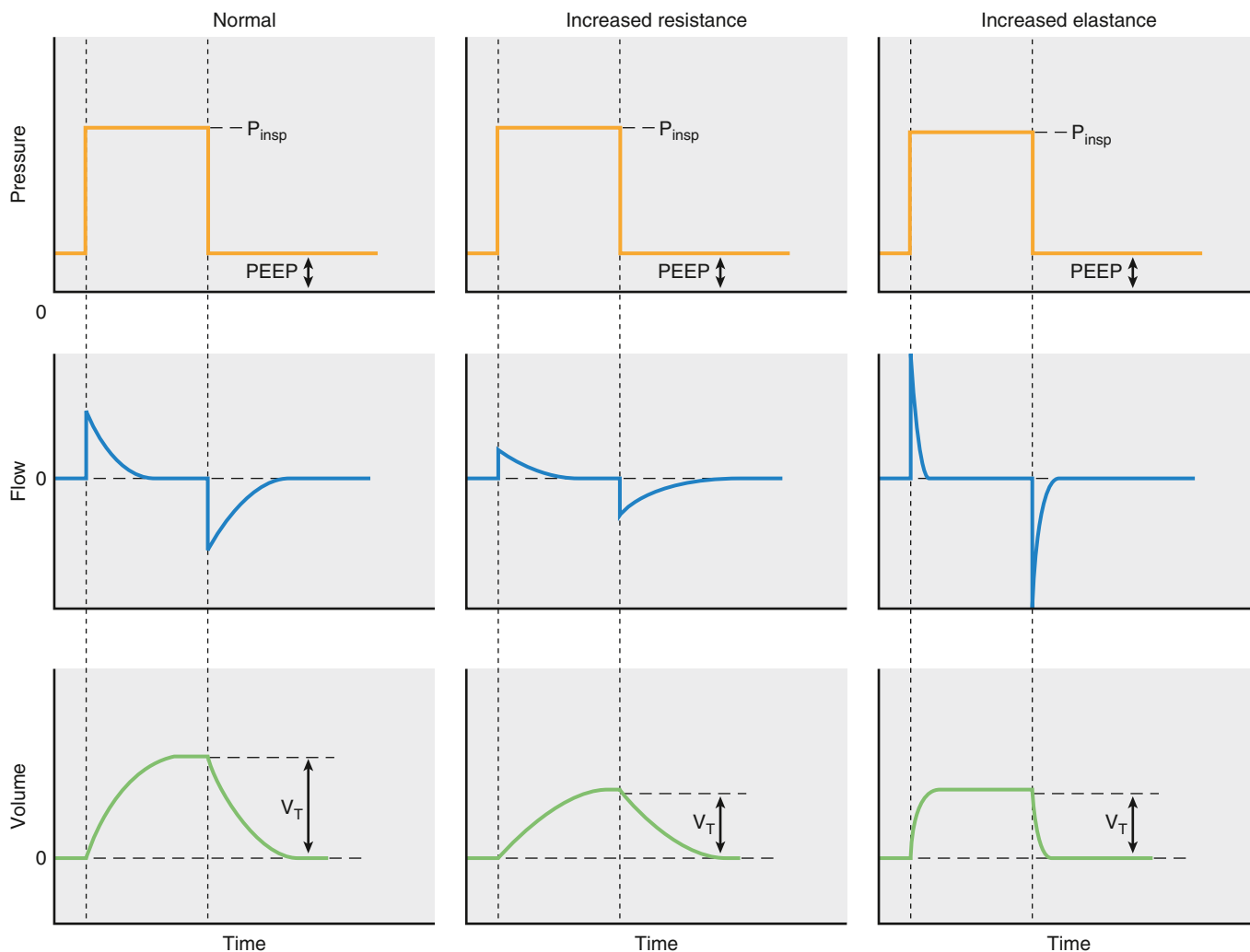


Fig. 37.21 Airway pressure, flow, and volume waveforms during pressure-controlled ventilation for the same patients depicted in Fig. 37.18. In this ventilatory mode, the ventilator delivers a fixed pressure during the inspiratory phase, irrespective of the resistance and compliance. Airflow and tidal volume will consequently change with changes in those parameters. Leftmost panels represent waveforms in a patient with normal lungs. Middle panels represent waveforms in a patient with airway obstruction. Note the absence of a peak-plateau difference as compared with the volume-controlled mode, a lower V_T , and a prolonged time for airflow to reach zero during exhalation as compared to the normal patient. Rightmost panels represent a subject with increased elastance, as may occur during laparoscopic insufflation or pneumothorax. Note a reduction in V_T and a shorter time to reach zero flow consistent with a more rigid lung. PEEP, positive end-expiratory pressure; P_{insp} , inspiratory pressure; V_T , tidal volume.

some patients because of dynamic airway compression or derecruitment.^{275,276}

Dynamic estimates of total respiratory R_{rs} and E_{rs} based on positive airway opening pressures are valid only during controlled modes of ventilation when the chest wall is relaxed. Although this is certainly the case during general anesthesia with neuromuscular blockade, the situation is more complicated during spontaneous breathing or assisted modes of ventilation. In such situations, the mechanics of the chest wall can be accounted for, if transpulmonary pressure is estimated using esophageal manometry, as described later.

STATIC RESPIRATORY MECHANICS

As previously illustrated, elastance (or compliance) is most easily measured during periods of zero flow because during these conditions the measured distending pressure is exclusively associated with elastic processes. However, a

more comprehensive characterization of elastance can be obtained by plotting cumulative inspired or expired lung volume against the distending pressure across the lungs or total respiratory system. This quasi-static PV curve can be constructed during very slow inflations or deflations²⁷⁷ (when resistive pressures are negligible), or during periodic flow occlusions.²⁷⁸ For example, the expiratory limb of the PV curve can be constructed by inflating the lungs to a specified volume (usually total lung capacity) and then allowing a passive deflation with flow interruptions of 1 to 2 seconds.

The quasi-static PV curves of the lungs or total respiratory system are inherently nonlinear (Fig. 37.22); that is, compliance, defined as the local slope (dP/dV) of the PV curve, varies with lung volume. Empirically, PV curves are described using single exponential²⁷⁹ or sigmoidal²⁸⁰ functions. When they are described using sigmoidal functions, two points are usually demarcated on such curves, which have traditionally been termed the upper inflection

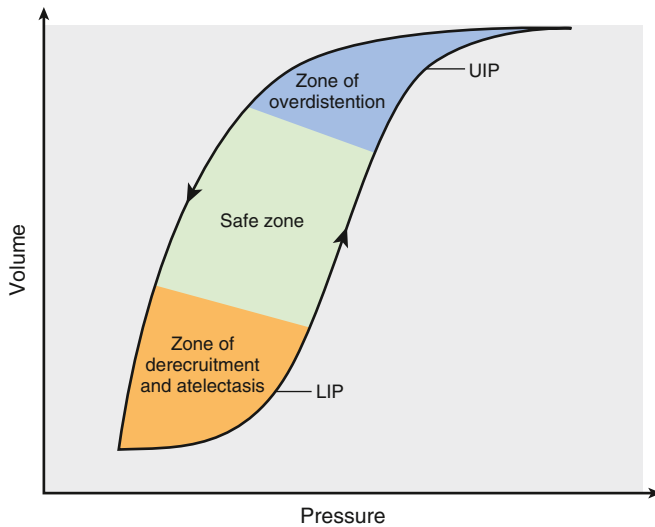


Fig. 37.22 Schematic of a pressure-volume curve for the lungs or total respiratory system, demonstrating hysteresis between inspiratory and expiratory limbs. Upper inflection point (UIP) and lower inflection point (LIP) are demarcated on the inspiratory limb. During mechanical ventilation, lung regions should be in the safe zone for optimal lung protection.

point (UIP) and lower inflection point (LIP).[‡] The UIP is thought to represent the point at which lung overdistention occurs, as during parenchymal strain-stiffening.²⁰³ The LIP is thought to reflect the process of maximum alveolar recruitment. During protective ventilation, one seeks to ventilate a patient in the most linear region of the PV curve.

Enough PEEP needs to be applied to avoid the LIP, where cyclic recruitment and derecruitment occur, along with lower V_{TS} to avoid the UIP and overdistention.²⁷⁸ Also, PV curves exhibit hysteresis, a phenomenon in which the lung volume at a specified pressure depends on the direction from which the distending pressure is approached (i.e., either inspiration or expiration). The reasons that the PV curve of the lungs or total respiratory system exhibits hysteresis are numerous and complex, and include the biophysical properties of surfactant,²⁸¹ time-dependent recruitment or derecruitment,²⁸² as well as contact friction among various connective tissue elements.²⁸³

There are a number of problems with managing ventilation based on quasi-static PV curves. First, PV curves are constructed only during periods of zero flow or near zero flow, and they do not reflect the mechanics of the lungs or total respiratory system during the dynamic processes of breathing or mechanical ventilation. In fact, the dynamic PV relationships for the same patient may be very different. Second, the UIP and LIP are not always visually apparent. Finally, a PV curve for the lungs or total respiratory system represents the averaged static behavior of a population of many lung units, with each operating on different portions of their own individual PV curves.

[‡]The formal mathematical definition of an inflection point is a point on a curve at which its concavity (i.e., its second derivative) changes sign from positive to negative or vice versa.

WORK OF BREATHING AND MECHANICAL POWER

The work of breathing (W) represents the energy required to inflate or deflate the lungs, or chest wall, or both, by a specified volume. In its simplest form, it can be represented as the product of the applied distending pressure and resulting volume change:

$$W = PV \quad (37.34)$$

However, when pressure and volume vary as functions of time (as during the course of a spontaneous or controlled breath), work may be expressed as the cumulative product²⁸⁴

$$W = \int PdV \quad (37.35)$$

or equivalently as the integral of the pressure-flow product with respect to time²⁸⁵

$$W = \int_{t=0}^T P(t) \dot{V}(t) dt \quad (37.36)$$

where T is the duration over which W is determined. For example, if T corresponds to the duration of inspiratory period and P the transpulmonary pressure, then W corresponds to the work required to overcome both the resistive and elastic forces opposing the entry of air into the lungs (Fig. 37.23). If T corresponds to the duration of an entire breath, the energy initially stored in the elastic recoil of the lung tissues will be recovered during a passive exhalation. In this case, the area of the resulting PV loop represents the work performed (i.e., energy lost) overcoming purely resistive losses in the airways and tissues. In general, W is assessed during inspiration only because the respiratory musculature (or a ventilator) must overcome both elastic and resistive pressures to bring air into the lungs. For a given V_T , W varies as a function of respiratory rate and, in most cases, achieves a minimum at a specified frequency. This frequency is termed the *energetically optimum breathing frequency*,²⁸⁵ as this is the rate at which energy expenditure is minimized. The derivative of W with respect to time is referred to as the *instantaneous mechanical power*, P ,

$$P = \frac{dW}{dt} \quad (37.37)$$

which, based on Eq. 37.31, can simply be expressed as the pressure-flow product with respect to time:

$$P(t) = P(t) \dot{V}(t) \quad (37.38)$$

Mechanical power, as an index of the *rate* of energy dissipation, can be used to assess the risk of developing ventilator-induced lung injury,²⁸⁶ particularly with changes in transpulmonary pressure during ventilation.²⁸⁷ Moreover, if the relationship between pressure and flow during ventilation can be described by relatively simple mathematical expressions (e.g., Eqs. 37.25–37.27), the mechanical power likewise can be embodied by simple analytic formulae.^{288,289} However, whether mechanical power may be predictive of ventilator-induced lung injury, or any meaningful clinical outcomes in ventilated patients with ARDS, is still a matter of contention.^{290–296}

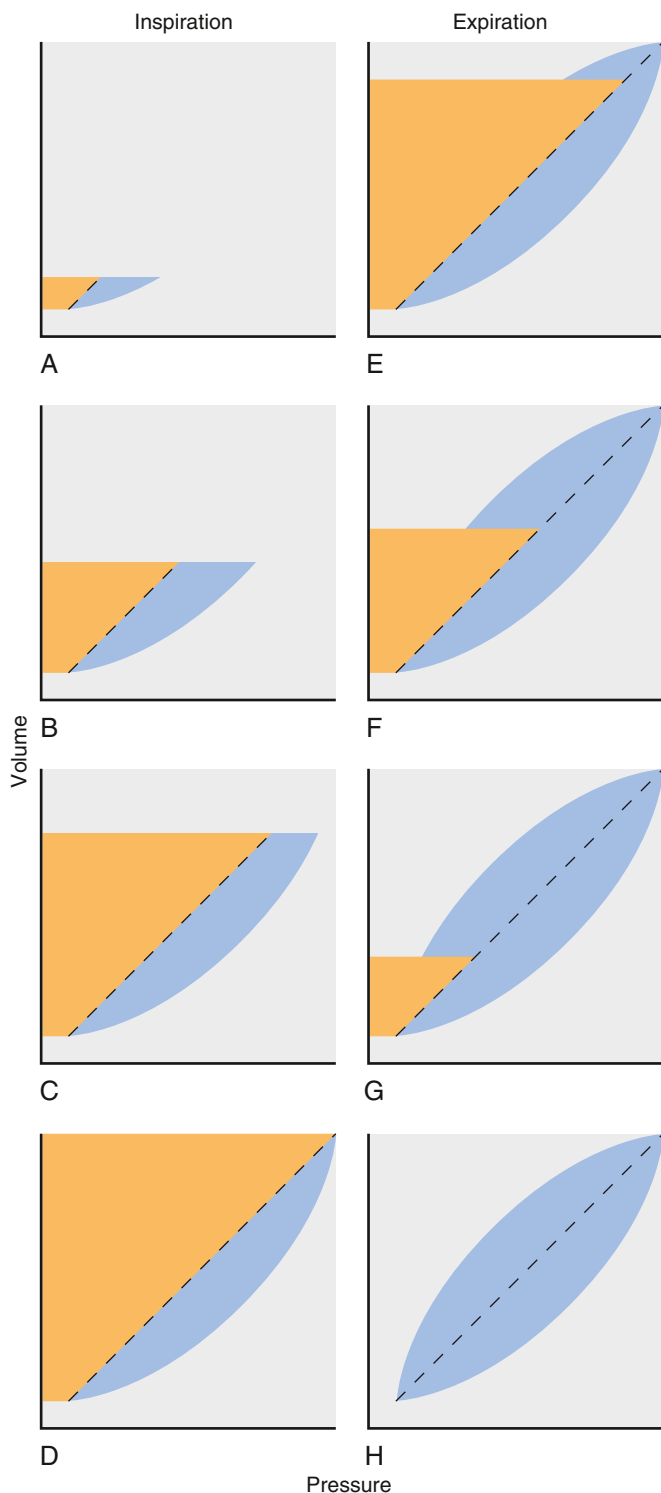


Fig. 37.23 Volume plotted against transpulmonary or (transrespiratory) pressure during the course of a single breath. Plots are shown during inspiration (panels A through D) and expiration (panels E through H). *Orange area* denotes the work of breathing due to tissue elasticity, while the *blue area* denotes the work of breathing due to airways and tissue resistance. Energy stored in elastic recoil at the end of inspiration will be entirely recovered during expiration. However, resistive energy will always be lost during both inspiration and expiration.

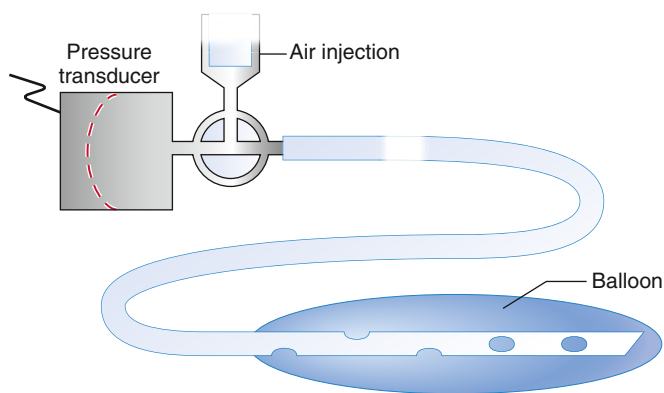


Fig. 37.24 Example of an esophageal balloon catheter. The proximal end of the catheter is connected to a pressure transducer, and air is injected into the balloon via a three-way stopcock. (Modified from Bates JHT. *Lung Mechanics: an Inverse Modeling Approach*. Cambridge: Cambridge University Press; 2009:220.)

MONITORING OF RESPIRATORY PRESSURES

Fundamental to any quantitative assessment of respiratory mechanics is the measurement of pressure. Such pressures may include those measured in the inspiratory or expiratory limb of the anesthesia machine, at the proximal end of an ETT, or within the trachea or esophagus. Pressure allows for the inference of the forces associated with the movement of gases through the airway tree, as well as the distention of the parenchymal tissues and chest wall, according to the elements described in Eq. 37.24. The most accessible and familiar pressure to the anesthesiologist or intensivist is the airway pressure during controlled mechanical ventilation. Ideally, this should be the pressure measured in the trachea or at the airway opening, exclusive of any distortions from airway devices or breathing circuits. However, for practical reasons, this “airway” pressure is not the actual pressure at the airway opening or within the trachea, but is the pressure transduced at the anesthesia machine or ventilator and reflects the resistive and compliant properties of the breathing circuit and face mask or ETT, as well as the mechanical properties of the respiratory system. Although many ventilators now use computer algorithms to provide some compensation for flow and pressure losses in the breathing circuit,²⁹⁷ such approaches often rely on idealized, linear models that do not truly reflect the complex processes of turbulent air flow, polytropic gas compression, or viscoelastic tubing walls.^{298,299} Therefore, one must use caution when making physiologic inferences based on airway pressures and volumes reported by a ventilator.

Often airway pressure is improperly used as a surrogate for lung distention. *Transrespiratory pressure* refers to the pressure drop across the lungs and chest wall, which during positive-pressure ventilation is usually determined by the difference between airway pressure and atmospheric pressure. Several processes that may contribute to increased transrespiratory pressure do not correspond to excessive lung inflation. For example, obesity, abdominal insufflation, or steep Trendelenburg position all may contribute to high airway opening pressures, but they do not necessarily indicate parenchymal overdistention.

By contrast, *transpulmonary pressure* refers to the distending pressure across the lungs alone. Its determination requires not only measurement of airway opening pressure, but also estimates of the pressure within the pleural space. This can be obtained relatively noninvasively using an esophageal balloon catheter, because the pressure measured within the esophagus is relatively close to intrapleural pressure.^{300,301} Such catheters are typically 100 cm long, with side holes in the distal tip covered by a thin-walled balloon (Fig. 37.24). The catheter can be placed through the mouth or nares and positioned in the middle-to-distal third of the esophagus. The catheter is connected to a transducer by a three-way stopcock, and a small amount of air is injected into the balloon such that its walls remain flaccid and do not contribute an additional recoil pressure to the measurement. Because the local values of pleural pressure vary depending on gravity, the balloon should be several centimeters in length to provide an average estimate of the pressure field surrounding the lungs. Estimating pleural pressure based on esophageal manometry has several limitations, including mediastinal compression of the balloon in supine patients, catheter migration, and cardiac artifact on the tracing.^{302,303} Nonetheless, its use in critically ill patients and under certain special conditions such as obesity, abdominal hypertension (primary or during laparoscopic insufflation), and extreme Trendelenburg position may be valuable in the adjustment of appropriate levels of PEEP.^{304,305} Peak and plateau pressures, which are typical in clinical practice, are obtained from the *transrespiratory* pressure. Accordingly, they do not characterize the pressures acting exclusively on the lungs, but on the whole respiratory system. Current recommendations suggest limiting transrespiratory plateau pressures to 26 to 30 cmH₂O, to minimize alveolar overdistention.³⁰⁶ However, such recommendations should be applied in appropriate clinical context, given that the same plateau pressure could correspond to very different transpulmonary pressures, and consequently to very different degrees of risk for lung injury, depending on the partitioning between lung and chest wall pressure components.^{307,308}

More recent studies suggest that the “driving pressure” – that is, the V_T normalized by the respiratory system compliance ($\Delta P = V_T/C_{RS}$) – may be one of the most important ventilatory variables to consider when stratifying mortality risk during ARDS.^{309,310} Its relationship to postoperative pulmonary complications in surgical patients has also been demonstrated.^{311,312} In practice, driving pressure may be computed as the difference between P_{plat} and PEEP.³⁰⁹

Auto-PEEP or *intrinsic-PEEP* is the positive pressure present within the alveoli at end-exhalation and is typically observed in ventilated patients with COPD, who demonstrate dynamic airway compression and expiratory flow limitation, as well as a significant portion of patients with ARDS, sepsis, and respiratory muscle weakness.³¹³ Auto-PEEP can promote significant respiratory and hemodynamic compromise. It is usually caused by a combination of increased airway resistance and reduced lung elastic recoil (i.e., an increase in the expiratory time constant of the respiratory system). Other factors are a decrease in expiratory time, an increase in V_T , an increase in external expiratory resistance, and persistent inspiratory muscle activity during exhalation. The amount of auto-PEEP can be estimated in mechanically ventilated patients by occluding the airway

at end exhalation, and observing a rise in airway pressure during the occlusion maneuver until a plateau is visible (<4 seconds) (Fig. 37.25). Auto-PEEP is then defined as the difference between end-occlusion to preocclusion airway pressures. Other methods can be used to assess auto-PEEP under dynamic conditions, as well as in spontaneously breathing patients with the use of an esophageal balloon.^{313,314}

Any measurement of pressure requires pressure transducers. Most pressure transducers are differential sensors with two input channels, and they produce an electrical output that is proportional to the pressure difference between these channels (Fig. 37.26A). Such transducers exhibit a high common-mode rejection ratio, defined as the tendency of the transducer to produce zero output if both of its inputs are exposed to identical pressures. Many pressure waveforms are transduced relative to atmosphere with one of the two inputs left opened to ambient air (Fig. 37.26B), otherwise referred to as gauge configuration. In clinical settings, pressure is most commonly measured using relatively inexpensive, piezoresistive transducers.²³⁸ These devices rely on a pressure-sensing diaphragm whose electrical resistance changes when it is deformed in response to a differential pressure. This change in electrical resistance can be sensed with a standard Wheatstone bridge circuit, which produces an output-voltage signal appropriate for amplification and filtering. In general, piezoresistive transducers have adequate frequency responses sufficient for most respiratory applications.^{315,316} However, this frequency response can be easily degraded if the transducer is connected to the pressure-sensing location with long, compliant tubing.³¹⁷

MONITORING OF RESPIRATORY FLOWS

Ideally, any measurement of flow should indicate the exact rate at which gas is entering and leaving the patient's lungs. This can be most easily accomplished by placing the flow-measuring device as close to the patient as possible, such as between the circuit Y-connector and the proximal end of the ETT or LMA. However, for practical reasons, most ventilators and anesthesia machines measure flow close to the machine console. This flow can be considerably different from that close to the patient because of the effects of gas compression, wall distention of the breathing circuit, and changes in gas humidity.

An important clinical use of flow monitoring is the detection of nonzero end-expiratory flows. This indicates that a gradient of pressure exists between alveolar regions and the upper airway with incomplete lung emptying at end-expiration, characteristic of auto-PEEP (P_{VolFI}-7). As a consequence, end-expiratory lung volumes would be larger than those in the absence of auto-PEEP, with risk of dynamic lung hyperinflation and reduction of venous return. Increase in expiratory time and reduction of V_T or respiratory rate should be considered to eliminate the issue.

During ventilation, airway flow can often be inferred according to time-dependent changes in delivered or inspired V_T or from the displacement waveform of a piston or cylinder arrangement or bellows.^{318–320} Preferably, flow is measured using transducers designed specifically for this task. The most common method involves the measurement of a differential pressure decrease (ΔP) across a fluid resistive element. Ideally, this ΔP is linearly related to the

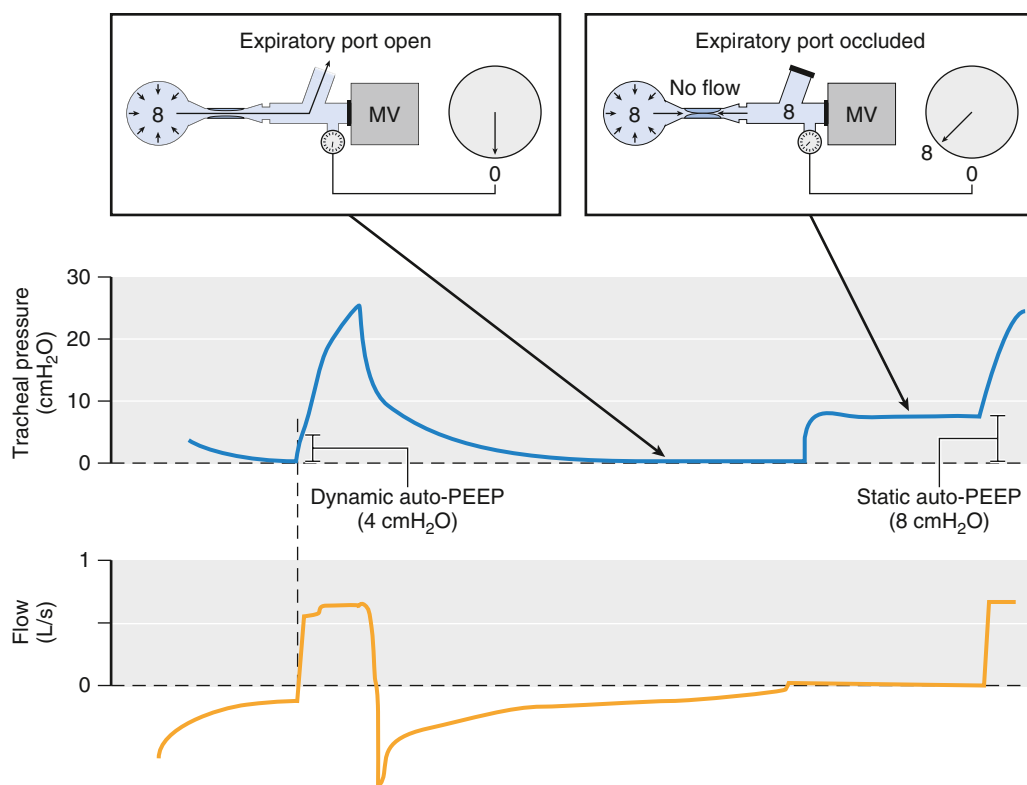


Fig. 37.25 Concept of intrinsic positive end-expiratory pressure (auto-PEEP). Shown are tracheal pressure and flow waveforms during controlled mechanical ventilation. Dynamic auto-PEEP can be estimated at the beginning of inspiration as the tracheal pressure when the flow trace equals zero. Static auto-PEEP can be estimated as the tracheal pressure following a prolonged expiratory occlusion. (Modified from Blanch L, Bernabé F, Lucangelo U. Measurement of air trapping, intrinsic positive end-expiratory pressure, and dynamic hyperinflation in mechanically ventilated patients. *Respir Care*. 2005;50:110–123; and Moon RE, Camporesi EM. Respiratory monitoring. In: Miller RD, Fleisher LA, Johns RA, eds. *Miller's Anesthesia*, 6th ed. New York, NY: Churchill Livingstone; 2005:1255, 1295.)

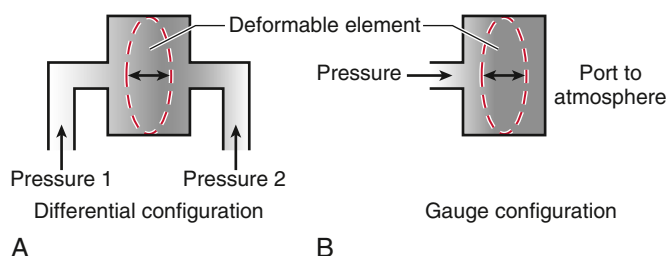


Fig. 37.26 Pressure transducers arranged in differential (A) or gauge (B) configurations. (Modified from Bates JHT. *Lung Mechanics: An Inverse Modeling Approach*. Cambridge: Cambridge University Press; 2009:220.)

corresponding flow through the device. Such a principle forms the basis for one of the oldest and most widely used flowmeters, the pneumotachograph. Although the instruments provide accurate measurement of flow, pneumotachographs are very sensitive to changes in temperature, humidity, and gas composition,³²¹ and require frequent calibration using various electronic or software-based techniques to ensure accurate measurements.^{322,323} This sensitivity, combined with difficulties in cleaning and sterilization, limits the routine use of these instruments in clinical environments. Orifice flowmeters also have the advantage of having relatively large internal diameters that limit the formation of condensation and minimize obstruction with secretions. Given that these devices can be manufactured using inexpensive plastic injection molding techniques,

many orifice flowmeters are designed to be disposable, thus making their use increasingly popular for respiratory applications in clinical environments.³²⁴

Most anesthesia machines use hot wire anemometers for flow measurement,³²⁵ which rely on temperature-dependent changes in the electrical resistance of a current carrying wire. When gas flows past the wire, the corresponding temperature drop changes the conductivity of the filament, which can be sensed with appropriate electronic circuits.³²⁶ Because a single wire filament cannot sense the direction of flow, two different anemometers can be used in the ventilator circuit: one for the inspiratory limb, and the other for the expiratory limb. Alternatively, two wires in series must be used for sensing bidirectional flow through a single conduit,³²⁷ with the flow direction determined according to which wire is cooled first. In general, hot wire anemometers have a better dynamic response compared with pneumotachographs or orifice flowmeters,³²⁸ making them ideal for flow measurement during HFV.^{270,329} However, these devices must also be calibrated to account for variations in gas density or humidity as well.³³⁰

MONITORING OF RESPIRATORY VOLUME

Similar to flow measurement, the ideal measurement of volume should be an accurate accounting of the gas quantity moving into and out of the patient. However, delivered or exhaled V_T measurement is subjected to the same

limitations of flow measurement because most anesthesia machines and ventilators determine volume by electrically or numerically integrating the corresponding flow signal.²³⁸ Care must be taken to ensure any drifts or offsets from the flow transducer are minimized, given that their integration will result in errors in estimated volume. This may be accomplished by appropriately zeroing the transducer periodically, such as at the end of a breath. Although electronic or digital high-pass filters may be used to remove drifts and offsets in real-time, these tend to have long transient responses.^{331,332}

Finally, besides the pathophysiologic and monitoring applications described in this section, monitoring of pressures, volumes, and flows has been applied to monitoring, training, and education for neonatal resuscitation.³³³

Plethysmographic Monitoring

Respiratory inductance plethysmography (RIP) is a non-invasive respiratory monitoring technique that quantifies changes in the cross-sectional area of the chest wall and the abdominal compartments. The method is used to assess V_T , respiratory rate, adequacy of high-frequency oscillatory ventilation (HFOV), lung volume changes during tracheobronchial suctioning, and thoracoabdominal synchrony.³³⁴ RIP relies on the principle that a current applied through a loop of wire generates a magnetic field normal to the orientation of the loop (Faraday's law) and that a change in the area enclosed by the loop creates an opposing current within the loop directly proportional to the change in the area (Lenz's law). Two elastic bands containing a conductor are used: one typically placed around the patient's chest, 3 cm above the xiphoid process; and the other around the abdomen. Each of these bands produces an independent signal representative of the thoracic and abdominal cross-sectional area, and the sum of these two signals calibrated against a known gas volume yields lung volume change.

There have been several pediatric applications of RIP to monitor V_T , respiratory rate, and breathing patterns aiming at postoperative apnea, building on the advantage that a face mask, LMA, or ETT is not required for measurements.^{335–338} The technique could also guide lung-protective ventilation strategies by facilitating the construction of PV curves for individual patients. These can be used to optimize lung recruitment, maintain an open lung, and limit overdistention (see the section on [Static Respiratory Mechanics](#), above). RIP has found several applications in sleep studies.

Intraoperatively, RIP may be useful in areas for which accurate monitoring of V_T is required but not possible with the usual anesthesia machine monitors. This could occur with a shared airway (e.g., laryngotracheal surgery,³³⁹ flexible and rigid bronchoscopy³⁴⁰), or a patient whose trachea is not intubated (e.g., monitored anesthesia care and noninvasive pressure support ventilation).

Limitations of intraoperative RIP include the fact that it cannot be used during thoracic and abdominal surgery, because of the need for the measurement bands. Also, the calibration characteristics of the device are affected by changes in the pattern of breathing.³⁴¹

Respiratory Rate Monitoring: Apnea Monitoring

Apnea and bradypnea are life-threatening events frequently observed in the intraoperative and postoperative periods of anesthesia. Prematurity, morbid obesity, age, obstructive sleep apnea, and central nervous system depressant medications have been associated with risk of apnea or bradypnea.^{342,343} Accordingly, many different approaches have been developed for detecting those events.^{344,345} Two main types of apnea are observed: central and obstructive. Central apnea is defined as apnea resulting from a failure of the central nervous system to drive respiration. Obstructive apnea is that resulting from upper airway obstruction. Current monitors assess at least one of three processes occurring during breathing to detect apnea³⁴⁵: chest wall expansion, gas flow, and gas exchange.

Chest wall expansion is usually measured as follows:

1. Changes in thoracic electrical impedance (impedance pneumography) of the chest wall. The method is based on the changes in electrical conductivity of the chest to an electrical current as air moves in and out of the lungs during breathing and blood volume changes in the same period. This is because air is a poor electrical conductor and blood is a good electrical conductor. A low electrical current at high frequency is applied in two chest electrodes, corresponding changes in voltage of the chest are measured, and the impedance is continuously calculated from these. The technique is implemented in several commercial systems using routine electrocardiographic leads and is also used in home monitoring of neonatal apnea.
2. Inductive plethysmography (as described earlier).
3. Abdominal and chest fiberoptic and resistive strain gauges (a pressure pad placed alongside the infant's rib cage, pneumatic abdominal sensors) are used.
4. Electromyographic signal of respiratory muscles is not frequently used because of the low signal-to-noise ratio.

An important drawback of techniques based on chest expansion is their inaccuracy in the presence of movement. Thus, obstructive apnea may be falsely assessed as normal respirations.³⁴⁶

Gas flow methods are based on measurements of different variables directly related to the presence of air flow in the airway:

1. Pressure gradients along the breathing circuit. This approach uses the Poiseuille principle ($\Delta P = k \times V$) and differential pressure transducers to detect flow.
2. Temperature of the breathed air in the nose or mouth is used.
3. Rapid response hygrometer is based on assessment of humidity in the exhaled air.

Techniques based on gas exchange focus essentially on exhaled CO_2 , as routinely used in intubated patients in the operating room. In nonintubated patients, the use of specifically designed cannulae combining DO_2 and breathed gas sampling facilitates the monitoring of exhaled CO_2 . Capnography allows for early detection of respiratory depression

before O₂ desaturation, particularly when supplemental O₂ is administered.^{347,348} Off-line or in-line infrared sensors are the most common technique. Respiratory rate measurements using these techniques were more accurate than those obtained with thoracic impedance tomography in extubated patients in postanesthesia care units, even with supplemental O₂ at high flows.³⁴⁹ Accurate end-tidal CO₂ measurements through a nasal cannula may be difficult in mouth breathers, including obese patients and those with obstructive sleep apnea. In such cases, a nasal cannula with an oral guide may improve measurement accuracy.³⁵⁰ Simulation studies indicated that respiratory rate can be accurately monitored during esophagogastroduodenoscopies using a CO₂ sampling bite block or a nasal cannula with oral cup. The accuracy of P_{ET}CO₂ measurements depends on the device used, O₂ flow, the intensity of oral breathing, and minute ventilation.³⁵¹

Pulse oximetry cannot be used as the primary monitor of apnea or bradypnea because O₂ desaturation occurs only late in a well-oxygenated apneic patient. However, pulse oximetry adds an additional level of safety combined with primary monitors of ventilation. For example, in a study using pulse oximetry and noninvasive capnography for continuous monitoring of 178 patients receiving patient-controlled analgesia, respiratory depression measured by O₂ desaturation occurred in 12% of the patients, in line with previous studies.³⁴² In contrast to previous estimates, episodes of bradypnea (respiratory rate <10 breaths/min) were present in 41% of patients, far more frequently than the previously reported 1% to 2%.^{352–354}

Respiratory rate monitoring is also critical to infant apnea monitoring. Combinations of transthoracic impedance and pulse oximetry may maximize the detection of true episodes of apnea in home monitors for neonatal apnea.³⁵⁵ This is because pulse oximetry provides an additional level of monitoring in cases of poor performance of transthoracic impedance secondary to motion artifacts. The value of combining monitors of ventilation such as capnography with pulse oximetry to maximize the detection of alveolar hypoventilation has been confirmed.³⁵⁶ Capnography allowed for early detection of arterial O₂ desaturation in cases of alveolar hypoventilation in the presence of supplemental O₂.

False-positive and false-negative alarms occur during apnea monitoring. The most dangerous conditions occur when artifacts are detected as breathing and alarms are not activated. Such artifacts include vibrations, cardiac motion, patient movement, and electromagnetic interference from other instrumentation. Impedance pneumographs are subject to cardiovascular artifacts, whereas pressure pads are subject to patient motion artifacts. A more common (yet less dangerous) problem is when an alarm sounds in a patient without apnea. Frequent causes include inadequate sensitivity settings, malfunctioning electrodes, and movement of the patient. Motion transducers based on sensing of acceleration were used in neonates undergoing HFOV, and results showed that monitoring of regional tidal displacement may enable the early recognition of deteriorating ventilation during HFOV that eventually leads to hypoxemia.³⁵⁷ This study found that in approximately half of the cases, hypoxemia had causes other than slowly deteriorating ventilation.³⁵⁷

Current recommendations of the Anesthesia Patient Safety Foundation for Essential Monitoring Strategies to Detect Clinically Significant Drug-Induced Respiratory Depression in the Postoperative Period (<http://www.apsf.org/initiatives.php>), including opioid-induced ventilatory impairment, state that continuous electronic monitoring of oxygenation and ventilation should be available and considered for all patients, not only those at risk for postoperative respiratory insufficiency. This is because selective monitoring is likely to miss respiratory depression in patients without risk factors. These recommendations stress that maintaining *status quo* while awaiting newer technology is not acceptable, and that intermittent “spot checks” of oxygenation (pulse oximetry) and ventilation (nursing assessment) are not adequate for reliably recognizing clinically significant evolving drug-induced respiratory depression. Continuous electronic monitoring should not replace traditional intermittent nursing assessment and vigilance. All patients should have oxygenation continuously monitored by pulse oximetry. Capnography or other monitoring modalities that measure ventilation and airflow are indicated when supplemental O₂ is needed to maintain acceptable O₂ saturations. The recommendations also call attention to threshold-based alarm limits on individual physiologic variables, which may result in failure to recognize early signs of progressive hypoventilation by being either insufficiently, or too, sensitive (excess false alarms).

Unexpected death has been associated with three patterns of respiratory depression.^{358–360} (1) Hyperventilation compensated respiratory distress (e.g., from sepsis, pulmonary embolus, or congestive heart failure): patients present initially with stable S_pO₂ and decreasing P_aCO₂ as metabolic acidosis develops with associated compensatory hyperventilation. High respiratory rate is typical of this pattern. Ultimately, a slow desaturation is followed by an abrupt fall in oxygenation as the ventilatory response to worsening acidosis fails. (2) Progressive unidirectional hypoventilation or CO₂ narcosis: frequently due to opioid or other sedative overdose, it is characterized by a rise in P_aCO₂ (and P_{ET}CO₂) initially because of decreased minute ventilation, often while S_pO₂ remains above 90%. (3) Sentinel rapid airflow/O₂ saturation reduction with precipitous S_pO₂ fall observed in patients with obstructive sleep apnea, who are dependent on the arousal state to maintain oxygenation. Sleep results in precipitous hypoxemia during apnea with potential sudden arrest.

Imaging for Respiratory Monitoring

Imaging as a monitoring technique provides tremendous insight into pulmonary structure, function, and inflammation in both health and disease.^{361–363} However, exposure to radiation and bulky devices has prevented its use at the bedside. Technologic advances have led to the introduction of new modalities with more compact equipment for clinical use. This may herald an important shift in respiratory monitoring toward the increased use of bedside imaging. Such improvements come with the advantages of less radiation exposure, noninvasiveness, and more detailed physiologic information.

CHEST RADIOGRAPHY

Chest radiography has been the traditional imaging method to assess intrathoracic conditions in the operating room, postoperative anesthesia care unit, and ICUs. Anesthesiologists should be familiar with basic radiologic findings representative of important pulmonary conditions such as interstitial infiltrates, hyperinflation, pneumothorax, pleural effusion, and consolidation. The image during chest radiography, as well as computed tomography, is based on the physical principle that X-rays reaching a detector (e.g., film) depend on tissue absorption, which varies linearly with tissue density. However, radiation exposure restricts frequency of use. Furthermore, technical difficulties limit image quality. These difficulties include reduced resolution caused by patient movement during acquisition, and image distortion related to proximity of the X-ray beam source to the film/detector and posterior position of the film cassette.

ULTRASONOGRAPHY

Lung ultrasonography is a technique increasingly used in the perioperative, critical care, and emergency settings in adults and children.^{364–367} Critical and systematic approaches have proven that important clinical information can be obtained with lung ultrasonography at levels comparable to computed tomography.³⁶⁷ Such features add to the advantages of practicality, low cost, and absence of radiation or other significant biologic side effects. Excellent reviews have been published, and the reader should refer to those for additional learning.^{366,367}

Lung ultrasonography has been applied successfully in the assessment of pneumothorax, interstitial syndrome (i.e., cardiogenic and permeability pulmonary edema), lung consolidation, and pleural effusion. Currently available multipurpose ultrasonography probes can be used for specific portions of the pulmonary exam according to their characteristics. For instance, the high-frequency (10–12 MHz) linear array probe allows for detailed examination of the pleura and superficial changes, such as pneumothorax. This probe's limitations include its size, which impedes access to larger areas of lung tissue because of the interference of the ribs, and high frequency, which limits assessment of deeper structures. A lower-frequency (1–5 MHz) probe provides improved depth penetration and is used to assess supradiaphragmatic structures (pleural space, lung). To optimize lung visualization with a single probe, frequently probes with emission frequencies of 5 to 7 MHz are used and are usually small with a tip shaped so that an acoustic window on the lung parenchyma can be obtained on intercostal spaces. The 5-MHz microconvex probe is frequently favored,^{364,365} although curvilinear and phased-array probes are also appropriate.³⁶⁶

When examining the patient, a methodical approach should be used to ensure comprehensive assessment of lung structure and function. Different approaches have been proposed for point-of-care lung ultrasound.^{366,368} A complete exam involves the bilateral assessment of the anterior, lateral, and posterior lung. Although protocols vary, in the supine patient, each hemithorax should be assessed in at least six zones during emergencies: two anterior (separated by the third intercostal space), two lateral, and two

posterior. In routine assessments, 8-zone and 12-zone protocols are most commonly used.^{366,368–370} Because of the significant acoustic impedance differences between air, tissue, and bone (ribs), thoracic ultrasound is based primarily on characteristic artifacts and not only on the visualization of anatomic structures.³⁷¹ Videos of 1 to 2 breaths should be acquired to take advantage of the lung dynamics in the findings. Machine settings should be optimized at the start of the examination.³⁶⁶ Usually, studies are initiated with the identification of the ribs as hyperechoic lines producing significant ultrasonography shadow (Fig. 37.27A), followed by visualization of the pleural line approximately 0.5 to 1.0 cm deeper and between the ribs. This is typically a bright, slightly curved line. These are major structures to be identified because many pathologies of relevance to the anesthesiologist affect their observed pattern. In the normal lung, lung sliding represents the movement of the visceral on the parietal pleura during respiration and is another key sonographic finding to be appreciated. The magnitude of the movement is larger in regions closer to the diaphragm than those near the lung apex.

Below the pleura, hyperechoic horizontal reverberation artifacts at regular distances are referred to as A-lines (see Fig. 37.27A). Another artifact is the B-line, a discrete laser-like vertical hyperechoic reverberation artifact that arises from the pleural line (previously described as “comet tails”), extends to the bottom of the screen without fading, moves synchronously with lung sliding, and erases A-lines (see Fig. 37.27B,C). Solitary B-lines are a normal finding in the normal lung, and increased numbers are observed in disease. Based on the recognition of such artifacts, pathologic conditions are identified.^{364–366,372} Most acute disorders involve the lung surface, and this explains the utility of thoracic ultrasound in visualizing them. Ultrasonography findings for pneumothorax are the absence of lung sliding, B-lines, and lung pulse, and the presence of lung points.

Pleural effusions are characterized by a usually anechoic space between the parietal and visceral pleurae (see Fig. 37.27C,F) and by respiratory movement of the lung within the effusion (“sinusoid sign”). The presence of echogenic material within the effusion suggests an exudate or hemorrhage, although some exudates are anechoic. Most transudates are anechoic. The M-mode ultrasound scan shows parallel lines indicative of no moving structure underlying the probe. The ultrasonographic finding designated as a lung point is found in the presence of pneumothorax and represents the imaging of the cyclic transition during breathing from the absence of any sliding or moving B-lines at a physical location (i.e., air with no lung parenchyma) to the visualization of lung sliding, B-lines, or altered A-lines indicative of lung tissue (see Fig. 37.27D).^{365,373} Bedside ultrasound scans can detect pneumothorax with sensitivity similar to a computed tomography scan.^{364,366}

An interstitial syndrome is characterized by the presence of multiple B-lines, a pattern potentially associated with the water/gas ratio in lung tissue.³⁷⁴ A positive region is defined by three or more B-lines in a longitudinal plane between two ribs (see Fig. 37.27E).³⁶⁵ Lung consolidation is characterized sonographically by a subpleural echo-poor region or one with tissue-like echo-texture (see Fig. 37.27F). Lung consolidations may be caused by infection, pulmonary embolism, lung cancer and metastasis,

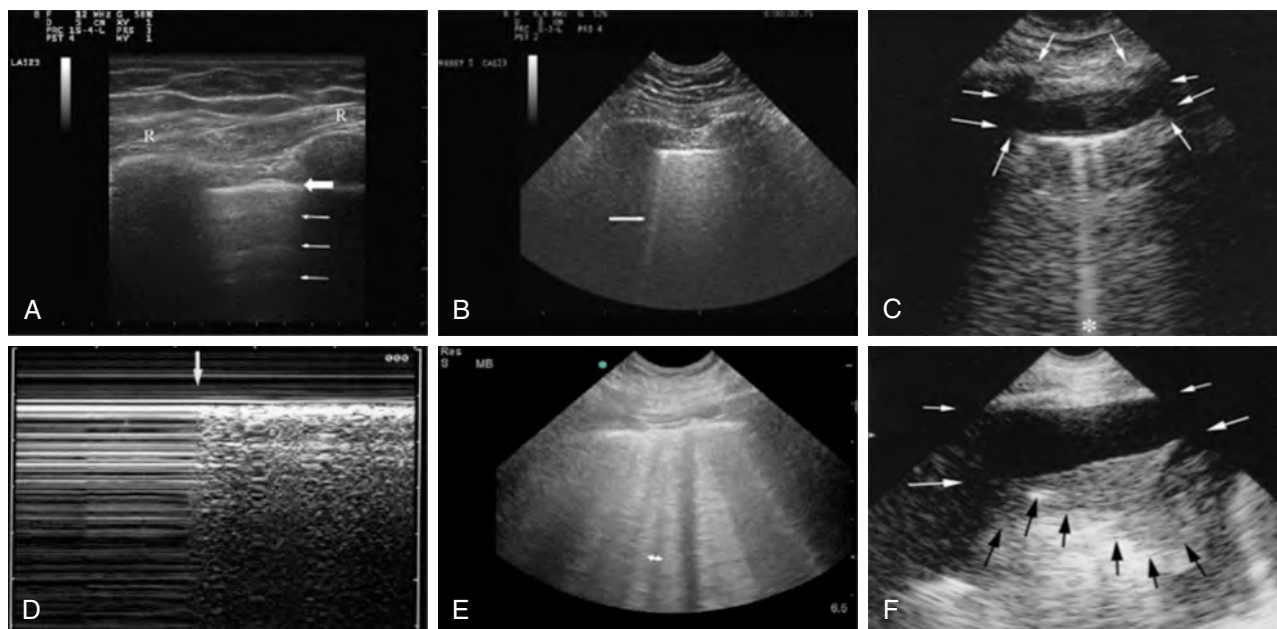


Fig. 37.27 (A) Typical thoracic view depicting adjacent ribs (R) producing acoustic shadowing. The white echogenic pleural line (block arrow) is approximately 0.5 cm below the ribs. A-line artifacts (line arrows) are seen at equidistant spaces below the pleural line. (B) B-line, or comet-tail artifact (line arrow), is the hyperechoic artifact extending from the pleural line to the edge of the screen, erasing the A-line. A solitary B-line is a usual finding in normal lungs. (C) Pleural effusion: image obtained at the posterolateral region showing the parietal pleura (upper horizontal arrows), the rib shadows (vertical arrows), and the visceral pleura (line indicated by the lower horizontal arrows) with the underlying lung. The dark anechoic region between the parietal and visceral pleura represents the pleural effusion. A B-line artifact (asterisk) originates from the lung and not from the pleural line, implying the presence of aeration. (D) M-mode of lung ultrasound illustrative of the “lung point” for diagnosis of pneumothorax. The sudden inspiratory transition from a parallel line pattern indicative of absence of lung motion (pneumothorax) to a granular pattern indicative of lung tissue can be observed (arrow). (E) Interstitial syndrome case showing B-lines 7 mm or less apart. Notice the increased number of B-lines as compared to panel (B). The pleural line (arrow) and ribs are also appreciated. (F) Lung consolidation with pleural effusion. As in (C), the image was obtained at the posterolateral region showing the parietal pleura (upper horizontal arrows) and the visceral pleura and lung (lower horizontal arrows) with an anechoic pleural effusion between them. Lung tissue denser than that presented in (C) with gas barriers (vertical arrows). This indicates pleural fluid with alveolar fluid often seen in critically ill patients. (A, B, D, E, From Turner JP, Dankoff J. Thoracic ultrasound. *Emerg Med Clin North Am.* 2012;30:451–473, ix. C, F, From Lichtenstein DA. BLUE-protocol and FALLS-protocol: two applications of lung ultrasound in the critically ill. *Chest.* 2015;147[6]:1659–1670.)

compression atelectasis, obstructive atelectasis, and lung contusion. Additional sonographic signs that may help determine the cause of lung consolidation include the quality of the deep margins of the consolidation, the presence of comet-tail reverberation artifacts at the far-field margin, the presence of air or fluid bronchograms, and the vascular pattern within the consolidation. Advances in clinical research and experience in lung ultrasonography allowed for the proposal of an algorithm to assess severe dyspnea in the acute setting (Fig. 37.28). The denominated BLUE protocol is a stepwise ultrasonographic approach to the patient with acute respiratory failure that aims at expeditious diagnosis with 90.5% accuracy.^{367,372,375} Several scoring systems have been developed to extend bedside lung ultrasound from a diagnostic to a monitoring modality.³⁶⁸

ELECTRICAL IMPEDANCE TOMOGRAPHY

Electrical impedance tomography (EIT) is a noninvasive and radiation-free imaging modality used to assess regional lung function at the bedside. The method is clinically available and has moderate-to-low spatial resolution but high temporal resolution, thus allowing for assessment of regional ventilation in real-time.^{270,363,376} Its use in estimating regional lung volume and optimizing mechanical ventilation settings brought significant attention to applications in ICUs and operating rooms.³⁶³

EIT is based on electrical impedance, a physical variable that reflects the opposition to the passage of a current

through an object when a voltage is applied across it.³⁷⁷ The impedance of biologic tissues depends on the tissue composition. High concentrations of electrolytes, extracellular water content, large cells, and number of cell connections by gap junctions as present in blood and muscles reduce impedance. Air, fat, and bone have high electrical impedance. Pathologic changes of tissue composition influence impedance. These include extravascular lung water (EVLW) (e.g., pulmonary edema), intrathoracic blood volume, fluid in cavities (pleural effusion, pericardial effusion, bronchial and alveolar fluid), foreign bodies (pleural drain), and lung fibrosis (e.g., after ARDs or as a primary disorder). During the breathing cycle, thoracic bioimpedance is influenced fundamentally by both ventilation and perfusion.

EIT relies on an array of electrodes (typically 16–32) around the chest region of interest. The location is chosen based on the clinical information desired, usually at the fifth intercostal space for standard lung assessment. The impedance information is provided in the form of a functional EIT image, or EIT waveforms that quantify changes in lung volume or perfusion in the studied thoracic cross section. Usually, images are presented with pixels representing relative impedance change, so-called functional EIT, because air corresponds to high impedances and fluid and tissue to low impedances, and the image represents regional lung ventilation. Absolute EIT (a-EIT) is the modality in which the image represents the actual impedance values. Direct assessment of lung conditions can be accomplished by comparing low impedance (e.g., hemothorax, pleural effusion, atelectasis,